

$$\mathcal{F}[x(t)](f) \equiv X(f)$$

$x(t)$ is real	$\Leftrightarrow X(-f) = X^*(f)$
$x(t)$ is imaginary	$\Leftrightarrow X(-f) = -X^*(f)$
$x(t)$ is even	$\Leftrightarrow X(f)$ is even
$x(t)$ is odd	$\Leftrightarrow X(f)$ is odd
$x(t)$ is real & even	$\Leftrightarrow X(f)$ is real & even
$x(t)$ is real & odd	$\Leftrightarrow X(f)$ is imaginary & odd
$x(t)$ is imaginary & even	$\Leftrightarrow X(f)$ is imaginary & even
$x(t)$ is imaginary & odd	$\Leftrightarrow X(f)$ is real & odd

Exercises

Prove the above

Fourier transform of even and odd functions

If $f(t) = f(-t)$

$$\int_{-\infty}^{\infty} f(t) \sin \omega t \, dt = 0$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} \, dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) \cos \omega t \, dt = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos \omega t \, dt$$

cosine transform, $\mathcal{F}_c(f)(\omega)$.

$$\mathcal{F}_c(f)(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos \omega t \, dt \equiv \hat{f}_c(\omega)$$

Since $\hat{f}_c(\omega) = \mathcal{F}_c(f)(\omega)$ is itself even the inverse transform is

$$\mathcal{F}_c^{-1}(\hat{f}_c)(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_c(\omega) \cos \omega t \, d\omega = f(t)$$

For $f(t) = -f(-t)$ (odd) functions

$$\int_{-\infty}^{\infty} f(t) \cos \omega t \, dt = 0$$

therefore we can use the *sine transform*

$$\hat{f}_s(\omega) = \mathcal{F}_s(f)(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin \omega t \, dt$$

and its invers

$$\mathcal{F}_s^{-1}(\hat{f}_s)(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}(\omega) \sin \omega t \, d\omega = f(t)$$

Example

Dirichlet function

$$f(t) = \begin{cases} 1, & |t| < 1 \\ 0, & |t| > 1 \\ \frac{1}{2}, & |t| = 1 \end{cases}$$

$$\hat{f}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^1 \cos \omega y \, dy = \sqrt{\frac{2}{\pi}} \frac{\sin \omega}{\omega} \equiv$$

$$f(t) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}(\omega) \cos \omega t \, d\omega$$

$$f(t) = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\sin \omega \cos \omega t}{\omega} \, d\omega = \begin{cases} 1, & |t| < 1 \\ 0, & |t| > 1 \\ \frac{1}{2}, & |t| = 1 \end{cases}$$

The integral representation of the function.

rect and normalized sinc functions

$$\text{rect}(t) = \begin{cases} 1, & |t| < 1/2 \\ 0, & |t| > 1/2 \\ \frac{1}{2}, & |t| = 1/2 \end{cases}$$

$$\text{sinc}(f) = \frac{\sin(\pi f)}{\pi f} = \mathcal{F}(\text{rect}(t))(f)$$

$$[f * g](t) = \int_{-\infty}^{\infty} f(t')g(t - t') dt'$$

$$[f * g](t) = [g * f](t) - \text{commutativity}$$

$$[f * g](t) = \mathcal{F}^{-1}[\hat{f} \cdot \hat{g}](t), \quad \mathcal{F}[f * g](\omega) = \hat{f}(\omega) \cdot \hat{g}(\omega)$$

Proof

$$\begin{aligned} f(t') &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega') e^{-i\omega' t'} d\omega' & g(t - t') &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{g}(\omega) e^{-i\omega(t-t')} d\omega. \\ [f * g](t) &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} dt' \int_{-\infty}^{\infty} d\omega' \hat{f}(\omega') e^{-i\omega' t'} \cdot \int_{-\infty}^{\infty} d\omega \hat{g}(\omega) e^{-i\omega(t-t')} = \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} dt' \int_{-\infty}^{\infty} d\omega' \int_{-\infty}^{\infty} d\omega \hat{f}(\omega') \cdot \hat{g}(\omega) \cdot e^{it'(\omega - \omega')} \cdot e^{-i\omega t} = \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' \int_{-\infty}^{\infty} d\omega \hat{f}(\omega') \hat{g}(\omega) \cdot \delta(\omega - \omega') \cdot e^{-i\omega t} = \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' \hat{f}(\omega') \cdot \hat{g}(\omega') e^{-i\omega' t} = \mathcal{F}^{-1}[\hat{f} \cdot \hat{g}](t) \end{aligned}$$

Exercises

Prove that

- 1 $[f * g](t) = [g * f](t)$ and that $\mathcal{F}[f * g](\omega) = \hat{f}(\omega) \cdot \hat{g}(\omega)$.
- 2 $f \cdot g = \mathcal{F}^{-1}[\hat{f} * \hat{g}]$
- 3 $f * \delta = f$
- 4 $f(t) * \delta(t - t_0) = f(t - t_0)$
- 5 $f * \Pi_T = T \sum_n f(t - nT)$

Example 1

Find the cosine and sine transform of $f(t) = e^{-at}$, $t \in (0, \infty)$; $a > 0$.

$$\text{Cosine transform} \rightarrow f(t) = \begin{cases} e^{-at}, & t > 0 \\ e^{at}, & t \leq 0; \end{cases}$$

$$\text{Sine transform} \rightarrow f(t) = \begin{cases} e^{-at}, & t > 0 \\ 0, & t = 0 \\ -e^{at}, & t < 0 \end{cases}$$

$$\hat{f}_C(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-at} \cos \omega t \, dt, \quad \hat{f}_S(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-at} \sin \omega t \, dt,$$

$$\hat{f}_C(\omega) + i\hat{f}_S(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-at} (\cos \omega t + i \sin \omega t) \, dt = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-(a-i\omega)t} \, dt =$$

$$= -\sqrt{\frac{2}{\pi}} \frac{1}{a-i\omega} \left[e^{-(a-i\omega)t} \right]_0^\infty = \sqrt{\frac{2}{\pi}} \frac{a+i\omega}{a^2+\omega^2}.$$

$$\rightarrow \hat{f}_C(\omega) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2+\omega^2}, \quad \hat{f}_S(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega}{a^2+\omega^2}.$$

By applying the invers transforms

$$\int_0^\infty \frac{\cos \omega t}{a^2+\omega^2} \, d\omega = \frac{\pi}{2a} e^{-at}, \quad \int_0^\infty \frac{\sin \omega t}{a^2+\omega^2} \, d\omega = \frac{\pi}{2} e^{-at},$$

we get the so called **Laplace integrals**.

Example 2

Solve the following integral equation:

$$\int_0^{\infty} g(u) \cos tu \, du = \phi(t), \text{ ahol } \phi(t) = \begin{cases} 1-t & \text{if } 0 < t \leq 1 \\ 0 & \text{if } t > 1. \end{cases}$$

It can be written also as

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} g(u) \cos tu \, du = f(t),$$

where

$$f(t) = \sqrt{\frac{2}{\pi}} \phi(t) = \begin{cases} \sqrt{\frac{2}{\pi}}(1-t) & \text{if } 0 < t \leq 1 \\ 0 & \text{if } t > 1. \end{cases}$$

Here $g(u)$ can be regarded as the cosine transform of $f(t)$ therefore

$$g(u) = \sqrt{\frac{2}{\pi}} \int_0^1 f(t) \cos ut \, dt + \sqrt{\frac{2}{\pi}} \int_1^{\infty} f(t) \cos ut \, dt.$$

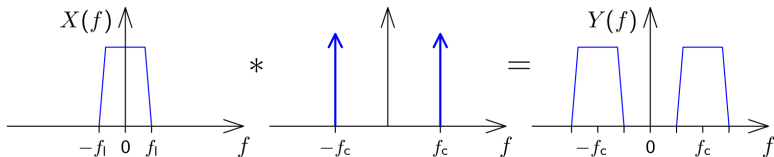
As $f(t) = 0$ if $t > 1$

$$g(u) = \frac{2}{\pi} \int_0^1 (1-t) \cos ut \, dt = \frac{2}{\pi} \frac{1 - \cos u}{u^2}.$$

Example: Amplitude modulation

If the frequency content of a signal $x(t)$ is in the $[-f_1, f_1]$ range that is not preferred for transmission (e.g., polluted by external effects) then we can shift it in the frequency domain into a range centered on f_c where it can get through unperturbed.

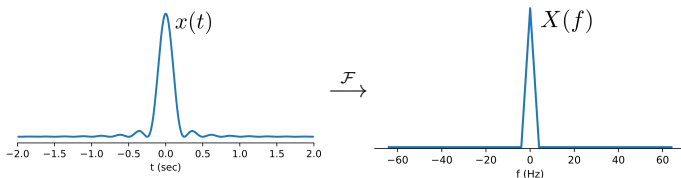
$$y(t) = A \cdot \cos(2\pi t f_c) \cdot x(t)$$



The spectrum of the baseband signal in the interval $-f_1 < f < f_1$ is shifted by the modulation to the intervals $\pm f_c - f_1 < f < \pm f_c + f_1$.

Sampling using a Dirac comb

Let's consider a signal $x(t)$ with its Fourier-transform $X(f) = \mathcal{F}(x(t))$:



The sampled signal is the discrete sequence

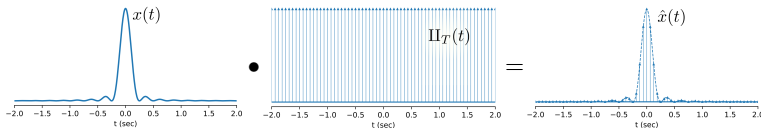
$$x[n] = x(nT) = x(n/f_s) = \int_{-\infty}^{\infty} x(\tau) \delta(nT - \tau) d\tau = x(t) * \delta(t - nT)$$

The continuous representation of the sampled signal is

$$\hat{x}(t) = x(t) \cdot \Pi_T(t)$$

where

$$\Pi_T(t) = T \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT)$$



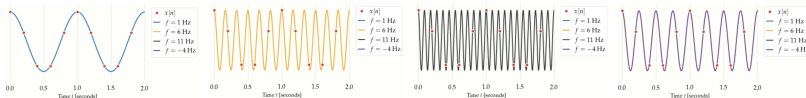
$\hat{x}(t)$ contains the same information as the discrete sequence $x[n]$, but is still in a form that can be analyzed using the Fourier-transform on continuous functions.

Questions:

- 1 Does $x[n]$ contain the same information as $x(t)$, i.e., the signal is reconstructible from its sample sequence?
- 2 If yes, under what conditions?
- 3 If yes and conditions are met how to do the reconstruction?
- 4 If not, what is the closest to full reconstruction that we can do?

Aliasing

- The time-continuous signal uniquely defines the sampled signal.
- The sampled signal does not uniquely define the original time-continuous signal.
- **Aliasing** is the phenomenon when two distinct continuous signals produce the same sequence of sample values when sampled at a fixed rate.



Consequence: high frequency signal will appear as low frequency if not sampled frequently enough!

Definition

Given a sampling rate f_s , two frequencies f and f' are aliases of each other if for some integer k

$$f' = f + kf_s$$

Theorem

For $x(t) = \cos(2\pi ft)$ and $x'(t) = \cos(2\pi f' t)$ if sampled with frequency $f_s \rightarrow x[n] = x'[n]$

Exercise

Prove the above theorem

Definitions

- A signal is **bandwidth limited** if for all harmonic components the corresponding frequencies $f \in [f_-, f_+]$;
- The **bandwidth** of such a signal is $f_+ - f_-$.

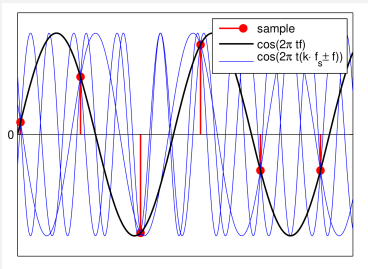
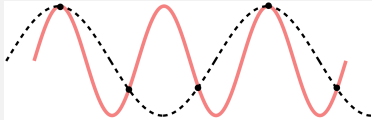
Observations

Support of a function $f(x)$ is defined as the domain wherein the function's value is different from zero:

$$\text{supp}(f) \equiv \{x \in \mathbb{R} | f(x) \neq 0\} \text{ .}$$

A signal $x(t)$ is bandwidth limited if its Fourier-transform, $X(f)$ has a finite support, i.e., $\text{supp}(X) = [f_-, f_+]$,
 $f_- < f_+ \in (-\infty, +\infty)$

Exercise



Write a short script demonstrating the effect of aliasing similarly to the figures above.

- **In theory:** if we know that the signal is bandwidth limited and $f_s > f_+ - f_-$, then for any f within the band $f \pm f_s$ will be outside of the band $\rightarrow$ aliasing is not an issue.
- **In practice:** since $\cos(2\pi ft) = \cos(2\pi(-f)t)$ a signal with frequencies in the $[f_-, f_+]$ band will have frequencies also in the $[-f_+, -f_-]$ band $\rightarrow$ the actual band is $[-f_+, f_+]$ $\rightarrow$ the actual bandwidth is $2f_+$.

Nyquist–Shannon sampling theorem

If a function $x(t)$ contains no frequencies higher than f , it is completely determined by giving its ordinates at a series of points spaced $1/(2f)$ seconds apart.

Or:

For a given sample rate f_s , perfect reconstruction is guaranteed possible for a bandlimit $f < f_s/2$.

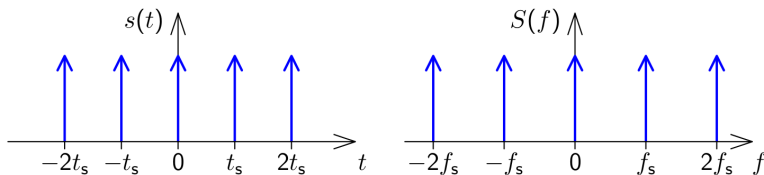
The threshold limit $f_s = 2f$ is called the **Nyquist-rate**.

We assume for now that the Nyquist-limit is respected at sampling, i.e., no frequencies higher than $f_s = 1/T$ are present in the signal.

Let's look at the spectrum of $\hat{x}(t)$.

The Fourier transform of a Dirac comb

$$\mathcal{F}(\Pi_T(t))(f) = \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = \frac{1}{f_s} \Pi_{f_s}(f)$$



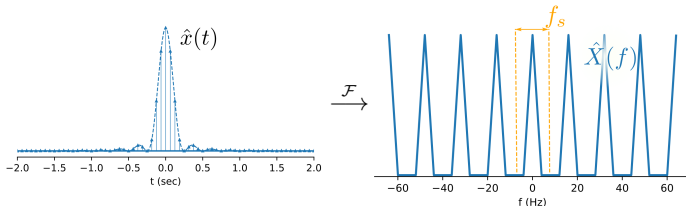
$$\hat{x}(t) = x(t) \cdot \Pi_T(t) = \mathcal{F}^{-1} \left[X * \frac{1}{f_s} \Pi_{f_s} \right]$$

$$\begin{aligned}
\hat{X}(f) &= \mathcal{F}[\hat{x}(t)](f) = \\
&= \mathcal{F}[x(t) \cdot \Pi_T(t)] = \\
&= X * \frac{1}{f_s} \Pi_{f_s} = \\
&= \frac{1}{f_s} \int_{-\infty}^{\infty} X(\nu) \Pi_{f_s}(f - \nu) d\nu = \\
&= \sum_{n=-\infty}^{\infty} X(f - nf_s)
\end{aligned}$$

Used formulae

$$\begin{aligned}
x \cdot y &= \mathcal{F}^{-1}[X * Y] \\
x * \Pi_T &= T \sum_n x(t - nT)
\end{aligned}$$

$$\mathcal{F}(\Pi_T(t))(f) = \frac{1}{f_s} \Pi_{f_s}(f)$$



The original signal's spectrum is repeatedly copied at f_s intervals. These are called **images** of the $X(f)$ spectrum.



sampling at $f_s > 2f_{\max}$ assures that the images do not overlap!!

Alternatively:

$$\begin{aligned}
 \hat{X}(f) &= \mathcal{F}[\hat{x}(t)](f) = \\
 &= \mathcal{F}[x(t) \cdot \Pi_T(t)] = \\
 &= T \cdot \int_{-\infty}^{+\infty} x(t) \sum_n \delta(t - nT) e^{2\pi i f t} dt = \\
 &= T \cdot \sum_n x[n] e^{2\pi i n \frac{f}{f_s}}
 \end{aligned}$$

which is the Fourier series expansion of an f_s periodic function where $T \cdot x[n]$ are the Fourier coefficients also defined as

$$x[n] = \int_{-f_s/2}^{f_s/2} \hat{X}(f) e^{-2\pi i n \frac{f}{f_s}} df$$

Indeed

$$\begin{aligned}
 \int_{-f_s/2}^{f_s/2} \hat{X}(f) e^{-2\pi i n \frac{f}{f_s}} df &= \int_{-\infty}^{\infty} \text{rect}(f/f_s) \cdot \hat{X}(f) e^{-2\pi i n \frac{f}{f_s}} df = \\
 &= \mathcal{F}^{-1}[\text{rect}(f/f_s) \cdot \hat{X}(f)](nT) = \left[\frac{1}{T} \text{sinc}_{\pi}(t/T) * \hat{x}(t) \right](nT) = \\
 &= \frac{1}{T} \int_{-\infty}^{\infty} x(\tau) \Pi_T(\tau) \text{sinc}_{\pi}\left(\frac{nT - \tau}{T}\right) d\tau = \sum_{m=-\infty}^{\infty} x[m] \text{sinc}_{\pi}(n - m) = \\
 &= \sum_{m=-\infty}^{\infty} x[m] \delta_{mn} = x[n]
 \end{aligned}$$

Used formulae

If $f(x)$ is 2ℓ periodic

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{n\pi x}{\ell}}.$$

$$c_n = \frac{1}{2\ell} \int_{-\ell}^{\ell} f(y) e^{-i \frac{n\pi y}{\ell}} dy$$

$$\text{rect}(t) = \begin{cases} 1, & |t| < 1/2 \\ 0, & |t| > 1/2 \\ \frac{1}{2}, & |t| = 1/2 \end{cases} = \mathcal{F}(\text{sinc}_{\pi}(f))(t)$$

$$\text{sinc}_{\pi}(f) = \frac{\sin(\pi f)}{\pi f} = \mathcal{F}(\text{rect}(t))(f)$$

$$\mathcal{F}[\text{rect}(t/T)](f) = \frac{1}{f_s} \text{sinc}_{\pi}(f/f_s) \text{ or}$$

$$\mathcal{F}[\text{rect}(f/f_s)](t) = \frac{1}{T} \text{sinc}_{\pi}(t/T)$$

$$\begin{aligned}
 x \cdot y &= \mathcal{F}^{-1}[X * Y] \\
 x * \Pi_T &= T \sum_n x(t - nT)
 \end{aligned}$$